Alphanomics Project 4

Ghadi Al Ghaddaf, Saskia Browning, Hanna Gos, Corentin Venet

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1 Full sample summary and report of BuyPCT / HoldPCT / SellPCT, Evaluation, and Incentives

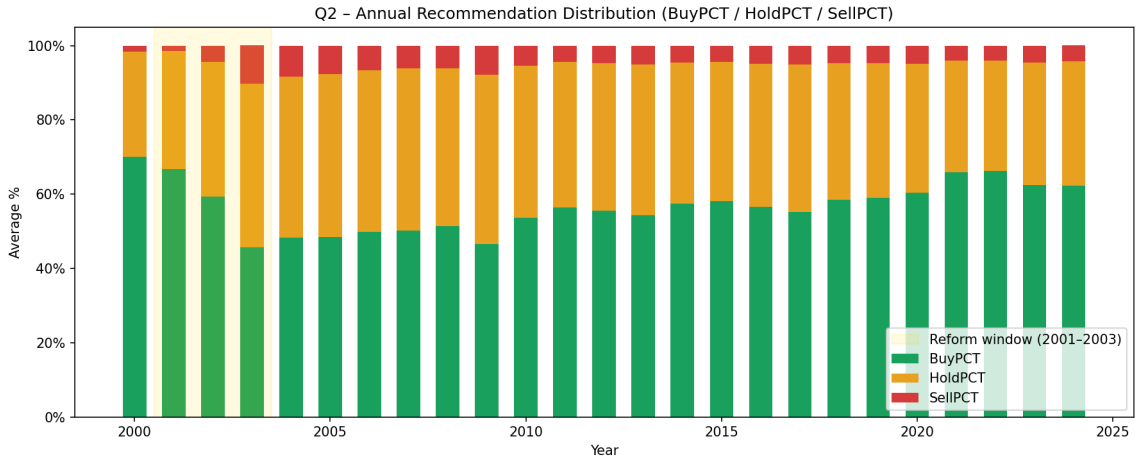
Table 1: Summary Statistics: Analyst Recommendation Frequencies (Full Sample)

	Mean (%)
Panel A: Average Recommendation Fractions	
Buy PCT	56.73
Hold PCT	38.08
Sell PCT	5.19

As the table above shows, over the sample period, the mean percentage of analysts issuing 'Buy' recommendations is approximately 57% versus 38% for 'Hold' and 5% for 'Sell'. There is a clear skew towards 'buy' and 'hold' recommendations - 11x the frequency of buy versus sell and 19x of these buy and hold combined versus sell. As discussed in class, there are blurred lines when it comes to analysts' incentives to rate stocks as 'buy' or even more broadly to just NOT rate them sell. There are conflicts of interest regarding the ratings of company stocks that the investment banking division of a bank has relationships with. The data we looked at in class showed that one of the key correlations to analyst compensation is investment banking 'contributions' – showing a p-value of ¡ 0.001 when regressing analyst compensation on a few variables. In this case, analysts are much more incentivised to treat these stock ratings more favourably or at least avoid more obvious negative signals like 'sell'.

2 Annual summary and report of BuyPCT / HoldPCT / SellPCT and regulation reliability

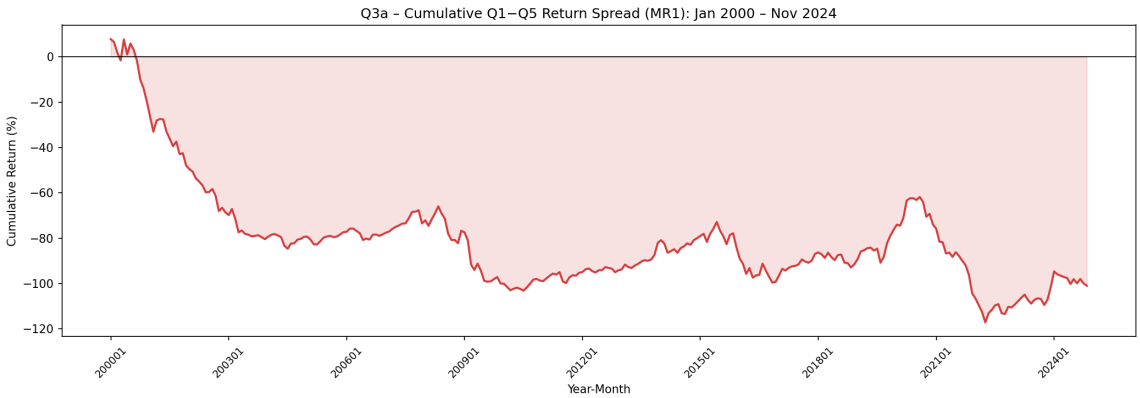
By looking at the average fraction of each type of rating by year, we can see a clear pattern whereby between 2000 and 2003, the % BuyPCT falls from an average of 70% to around 45%. Simultaneously, HoldPCT rises from 28% to 44% and SellPCT rises from 2% to 10%. In this way, it seems the regulation did indeed dissuade analysts from ranking as many stocks as 'buy' – the buy/sell ratio is 4.5x (less than half the total period average). However, one could argue that, if it's more just the lack of a 'sell' rating that's really important, the difference is less clear and only fell slightly because the cumulative buy + hold % is still 90% and the combined ratio to sell is 9x.In this case, perhaps to maintain relationships with the investment banking division, the more important thing was not rating these stocks 'sell'.



Either way, it kept 'buy' fraction hovering around 45-50% until 2009 when the fraction went back up and came back up to above 60%.

3 Portfolio formation based on analyst recommendations - one-month-ahead returns

a)



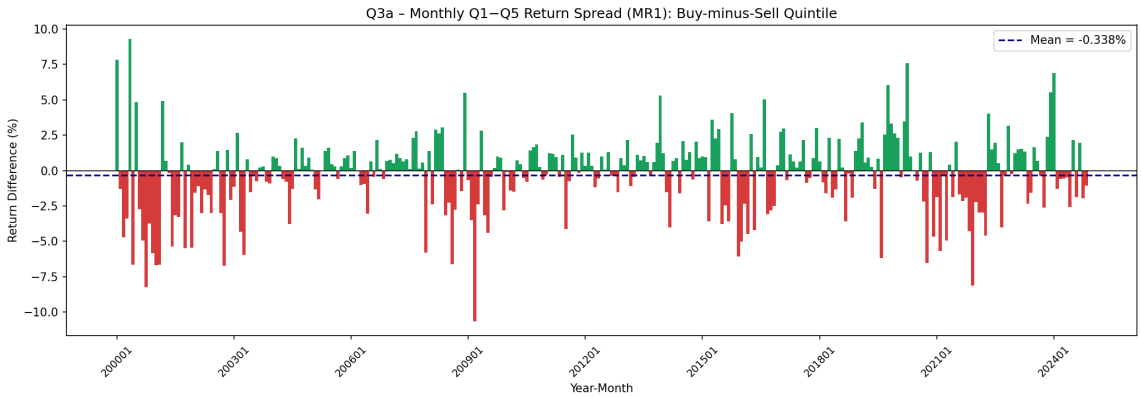
The monthly spread is defined as the average return of Q1 (most favorable recommendations) minus the average return of Q5 (least favorable recommendations). Economically, this is the return to a zero-cost, monthly rebalanced long-short strategy: long stocks with the strongest buy signals and short stocks with the strongest sell signals. If analyst recommendations contain positive forward-looking information, this spread should be consistently positive.

Empirically, the mean monthly spread is negative (-0.3379%), and the cumulative spread trends downward over the sample. This means that, on average, stocks in the "buy" bucket underperform

stocks in the "sell" bucket one month ahead after market adjustment. In other words, the sign of the spread goes against the intended predictive direction of analyst recommendations.

The month-to-month distribution is still mixed: 143 of 299 months are negative (47.8%) and 156 are positive (52.2%). So the key result is not that negative months dominate in count, but that the negative episodes are large enough to pull the average below zero. This distinction matters for interpretation: recommendations are not uniformly wrong every month, but they do not deliver reliable positive alpha in the direction they are supposed to.

Relative to the assignment objective, this time-series directly measures whether recommendations improve portfolio allocation in real time. A persistent positive spread would indicate useful signal value; the observed negative average spread instead suggests limited efficacy for a recommendation-following strategy in this sample period.



b)

It means the analysts' recommendations are pretty useless in informing the portfolio in the way they should but one might want to use their recommendations as a contrarian indicator i.e. - do the opposite of what they suggest since they seem to have negative predictive power over future returns.

If we take into account transaction costs and bid-ask spreads, this could fairly easily wipe out any marginal gains achieved from doing this contrarian strategy. Monthly rebalancing is expensive especially for smaller and less liquid stocks too so using them as a contrarian signal may not be profitable once accounting for the frictions.

c)

Table 2: Q3c – t-statistic: Q1–Q5 MR1 Spread

Metric	**Value**
N (months)	299
Mean return	-0.337852%
Std dev	2.774632%
Std error	0.160461%
t-statistic	-2.1055
Critical value (5%)	±1.96
Reject H_0?	YES

The table above shows us that this Q1 - Q5 strategy following the analysts' recommendations for strong buy and sell, with a mean of -0.338%, is statistically different from zero but in the negative direction, at the 5% significance level (the t-statistic is -2.1055 which is lower than the critical value -1.96 in the negative tail). Thus, we reject the null hypothesis of the mean return of this strategy being zero. Not only does the strategy not generate alpha but it actively loses money in a statistically significant way. Thus, it is actually a reliable contrarian signal.

4 Portfolio formation based on analyst recommendations - 12-month lagged return

Table 3: Average retlag12 by recommendation quintile

Quintile	**Average retlag12**
Q1 (Buy)	0.1643
Q2	0.1098
Q3	0.0636
Q4	-0.0035
Q5 (Sell)	-0.0423
Q1 – Q5 spread	0.2021

Table 4: Spread significance for retlag12 (Q1–Q5)

Metric	**Value**
N (months)	299
Mean spread	0.2021
Std dev	0.1562
Std error	0.0090
t-statistic	22.3809
Critical value (5%)	± 1.96
Reject H_0?	YES

Replacing MR1 with retlag12 reveals a strong monotonic pattern: stocks in better recommendation quintiles have substantially better trailing 12-month returns. The Q1–Q5 gap is large and highly significant, indicating recommendations are strongly aligned with *past* performance.

This supports the critique that sell-side recommendations often reflect momentum (what has already happened) more than they forecast future abnormal returns (what will happen next). In other words, analysts appear better at classifying recent winners versus losers than at generating actionable forward-looking alpha.